Unit testing report

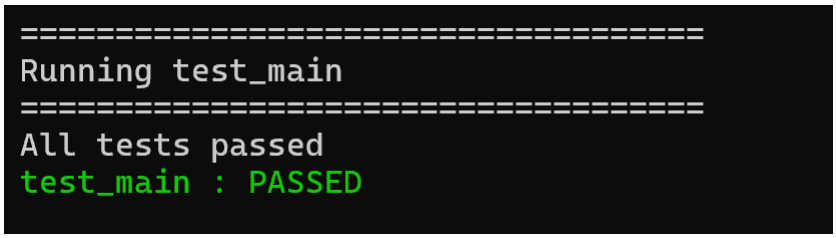
The purpose of unit testing is to verify that individual modules of the Online Quiz System function correctly and reliably. The testing was performed on the core application modules using C++ test files.

The tests focused on general quiz logic, administrator validation, question loading, and score reliability. Each test module was compiled and executed independently using the GCC compiler.

Component	**Details**
Project	Online Quiz System
Programming Language	C++
Compiler	GCC (g++)
Standard	C++17
Operating System	Windows
Testing Method	Unit Testing
Test Framework	C++ assertions and custom test programs
Test Execution Environment	PowerShell

The test files were compiled using the C++17 standard with warning flags enabled (-Wall -Wextra).

1)



Objective: To verify the general quiz logic and ensure that the main test cases execute successfully.

Expected Result: All test cases should execute successfully without assertion failures.

Actual Result: The terminal displayed "All tests passed".

Status: PASS

2)

```
==================================
Running test_admin_validation
==================================
QZ-18 and QZ-16 validation tests passed!
test_admin_validation : PASSED
```

Objective: To verify administrator validation functionality.

Test Cases:

- QZ-18: Administrator validation.
- QZ-16: Administrator validation.

Expected Result: Both validation tests should pass.

Actual Result: The terminal displayed "QZ-18 and QZ-16 validation tests passed!"

Status: PASS

3)

```
Warning: Skipping invalid question at line 5.
Warning: Invalid topic at line 6.
Warning: Skipping malformed record at line 7. Expected seven fields.
Warning: Skipping malformed record at line 3. Expected seven fields.
Warning: Invalid answer at line 4.
Warning: Skipping invalid question at line 5.
Warning: Invalid topic at line 6.
Warning: Skipping malformed record at line 7. Expected seven fields.
Question loader tests passed!
test_question_loader : PASSED
```

Objective: To verify that the question loader correctly processes valid question records and rejects invalid records.The following conditions were tested:

1. Loading valid questions.
2. Handling malformed records.
3. Rejecting invalid answer formats.
4. Rejecting questions with empty question text.
5. Rejecting invalid topics.
6. Verifying the number of successfully loaded questions.

Expected Result: Valid questions should be loaded successfully, while malformed or invalidrecords should be rejected.

Actual Result: The terminal displayed warning messages for invalid records and successfully loaded the two valid questions. The assertion checks passed, and the terminal displayed "Question loader tests passed!"

Status: PASS

4)

```
====================================
Running test_score_reliability
====================================
Run 1: score = 3
Run 2: score = 3
Run 3: score = 3
QZ-20 score reliability tests passed
test_score_reliability : PASSED
```

Objective: To verify that the quiz scoring mechanism produces consistent results across repeated executions.

Test Execution:

Execution	**Score**
Run 1	3
Run 2	3
Run 3	3

Expected Result: The scoring mechanism should produce consistent scores across repeated executions using the same test conditions.

Actual Result: All three executions produced a score of 3. The terminal displayed "QZ-20 score reliability tests passed".

Status: PASS

```
========== FINAL RESULTS ==========
PS C:\Users\Hp.CHINMAY\Desktop\Quiz-system> Write-Host "Test suites passed: $passed"
Test suites passed: 4
PS C:\Users\Hp.CHINMAY\Desktop\Quiz-system> Write-Host "Test suites failed: $failed"
Test suites failed: 0
```

5) branch coverage



Observations

1. All four unit-testing suites passed.
2. The question loader correctly identified malformed records and invalid input.
3. Expected validation warnings were displayed during negative testing.
4. The score reliability test produced identical scores across three executions.
5. No failing test suites were reported in the final execution summary.